

Streamlit Frontend Interface Report (Optional Task)

Dual-Mode Web Application Implementation, Code Base & Expected Outputs

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Domain:	AIML (Artificial Intelligence and Machine Learning)
Technologies Used:	Streamlit, LangChain, Hugging Face Hub, Python

Git Hub Link : <https://github.com/omkulawade03/Frontend-Session-Assignments/tree/main/task31/Optional%20Task>

1. Task Overview & Architecture

The optional task focuses on delivering an interactive web user interface for the Dual-Mode Agentic AI Chatbot using Streamlit. It allows users to dynamically switch model backends between Cloud API Inference (DeepSeek-R1) and Local Pipeline Inference (TinyLlama-1.1B) via a sidebar configuration, maintaining clean session state management and real-time response rendering.

2. Source Code Implementation (app.py)

```
import os
import streamlit as st
from dotenv import load_dotenv
from langchain_core.prompts import ChatPromptTemplate
from langchain_huggingface import ChatHuggingFace, HuggingFaceEndpoint, HuggingFacePipeline

# Load environment variables
load_dotenv()

# Streamlit Page Configuration
st.set_page_config(page_title="Dual-Mode Agentic AI", page_icon="🤖", layout="centered")

st.title("🤖 Dual-Mode Agentic AI Chatbot")
st.caption("Powered by LangChain & Hugging Face")

# Sidebar for Model Selection
st.sidebar.header("Settings")
mode = st.sidebar.radio(
    "Select Model Mode:",
    ["API Mode (DeepSeek-R1)", "Local Mode (TinyLlama)"]
)

# Resource Caching for Efficiency
@st.cache_resource
```

```

def load_api_model():
    token = os.getenv("HUGGINGFACEHUB_API_TOKEN")
    if not token:
        return None
    llm = HuggingFaceEndpoint(
        repo_id="deepseek-ai/DeepSeek-R1",
        task="text-generation",
        max_new_tokens=256,
        temperature=0.7,
        huggingfacehub_api_token=token,
    )
    return ChatHuggingFace(llm=llm)

@st.cache_resource
def load_local_model():
    llm = HuggingFacePipeline.from_model_id(
        model_id="TinyLlama/TinyLlama-1.1B-Chat-v1.0",
        task="text-generation",
        pipeline_kwargs={"max_new_tokens": 150, "do_sample": True, "temperature": 0.7},
    )
    return ChatHuggingFace(llm=llm)

# Load selected model
if mode == "API Mode (DeepSeek-R1)":
    chat_model = load_api_model()
    if chat_model is None:
        st.error("Error: HUGGINGFACEHUB_API_TOKEN is missing in your .env file!")
        st.stop()
else:
    with st.spinner("Loading local model into memory..."):
        chat_model = load_local_model()

# Session State for Conversation History
if "messages" not in st.session_state:
    st.session_state.messages = []

# Render Previous Chat History
for message in st.session_state.messages:
    with st.chat_message(message["role"]):
        st.markdown(message["content"])

# Process User Chat Input
if prompt_input := st.chat_input("Type your message..."):
    st.session_state.messages.append({"role": "user", "content": prompt_input})
    with st.chat_message("user"):
        st.markdown(prompt_input)

    prompt_template = ChatPromptTemplate.from_messages([
        ("system", "You are a very polite, kind, and helpful AI assistant."),
        ("user", "{input}")
    ])
    chain = prompt_template | chat_model

    with st.chat_message("assistant"):
        with st.spinner("Thinking..."):

```

```

try:
    response = chain.invoke({"input": prompt_input})
    st.markdown(response.content)
    st.session_state.messages.append({"role": "assistant", "content":
response.content})
except Exception as e:
    st.error(f"Error generating response: {e}")

```

3. Execution Command & Terminal Output

To run the Streamlit application, execute the following command in your virtual environment:

```
uv run streamlit run app.py
```

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>

Network URL: <http://192.168.1.5:8501>

4. Expected Web Interface Output & Visual Layout

```

=====
Sidebar Panel [Settings]:
-----

```

- Model Radio Selector:
 - (•) API Mode (DeepSeek-R1)
 - () Local Mode (TinyLlama)

Main Panel Area:

```

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🤖 Dual-Mode Agentic AI Chatbot
Powered by LangChain & Hugging Face

```

[Chat Area - Interaction Example]

```

👤 User
Hi, introduce yourself in two sentences.

```

```

🤖 Assistant
Hello! I am a polite and helpful AI assistant powered by DeepSeek-R1, designed to help you with reasoning, coding, and general queries. It is a pleasure to meet you, and I look forward to assisting you today!

```

```

[Bottom Pinned Input Box]
[ Type your message... ]
=====

```